

Financial Calculation Analysis & Improvement Roadmap — 2026-08-18

Portfolio Manager v2. Compares current backend calculations against professional trading platforms (Interactive Brokers, Bloomberg Terminal, TradingView, Wealthsimple Trade, TD Direct Investing) and identifies gaps, algorithmic issues, and improvement opportunities.

Current Calculation Inventory (What Is Already Good)

Calculation	Implementation	Notes	
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RSI(14)	Wilder's Smoothed	Correct per TradingView standard	
RSI Signal (9-EMA)	EMA of full RSI series	Good leading indicator	
MACD (12,26,9)	Full histogram + slope detection	Above average — histogram delta is a real edge	
ATR(14)	Wilder's Smoothed	Correct	
Bollinger Bands (20, $\pm 2\sigma$)	Standard	Correct	
Stochastic %K (14)	Fast (raw) %K	Partially complete — see Issue 3	
Fibonacci Retracement	Swing high/low, 4 levels	Good zone classification	
Portfolio Beta	Weighted average, sector fallback	See Issue 7	
Value Screener (5-factor)	EBIT/EV, FCF, P/B, Piotroski, ROIC Solid engine — see Issues 8–12		
ATR Stop Loss	1.5× ATR from extreme	Good dynamic stop methodology	
Volume Ratio	Current / 20-day SMA	Acceptable	
EOD Volume Projection	Linear scaling with 2× cap	See Issue 14	
Portfolio Value History	Daily EOD snapshot	Good foundation for performance analytics	

Issue 1 — No Portfolio Performance Metrics Analysis Complete

****What is currently calculated****: Simple unrealized gain = Market Value – (Average Cost × Shares).

****What professional platforms provide****:

- ****Time-Weighted Return (TWRR)**** — Industry standard. Eliminates distortion from cash flows. Used by every Canadian broker (TD, RBC, Questrade) in performance reporting. Formula chains sub-period returns: $TWRR = [(1+R_1) \times (1+R_2) \times \dots (1+R_n)] - 1$.
- ****Money-Weighted Return (MWRR/IRR)**** — Accounts for the timing and size of cash flows. Better for personal performance measurement.
- ****Annualized Return**** — $(1 + \text{totalReturn})^{(365/\text{days})} - 1$ for cross-period comparisons.
- ****Sharpe Ratio**** — Risk-adjusted return: $(\text{Portfolio Return} - \text{Risk-Free Rate}) / \text{StdDev}(\text{Returns})$. 1.0+ is considered good.
- ****Sortino Ratio**** — Like Sharpe but penalises only downside volatility: $(\text{Portfolio Return} - \text{Risk-Free Rate}) / \text{StdDev}(\text{Negative Returns})$.
- ****Maximum Drawdown**** — Largest peak-to-trough decline. Critical risk assessment metric.
- ****Calmar Ratio**** — $\text{Annual Return} / \text{Max Drawdown}$. Efficiency of risk-taking.

****Current gap****: `PortfolioValueHistory`` table already captures daily total values — the correct data foundation. The analytics layer is missing.

****Recommendation****: Add a `PerformanceAnalyticsService`` that computes TWRR, Sharpe, max drawdown, and annualized return from `PortfolioValueHistories``. The data is already there.

****Effort: Medium.****

Issue 2 — No Risk-Adjusted Metrics Analysis Complete

****Missing metrics**:**

- ****Value at Risk (VaR)**** — 95%/99% confidence interval loss estimate. Standard in institutional platforms.
- ****Alpha**** — $\text{Portfolio Return} - (\text{RiskFreeRate} + \text{Beta} \times (\text{Market Return} - \text{RiskFreeRate}))$. Answers whether skill or market beta drove returns.
- ****Tracking Error**** — Standard deviation of return differences vs. benchmark.
- ****Correlation Matrix**** — Pairwise correlation between positions. Two correlated positions provide no true diversification benefit.

****Current state****: Beta is calculated (weighted average) but unused for return attribution or risk analysis.

****Recommendation****: Start with correlation matrix (uses historical price data already fetched) and max drawdown. VaR requires 30+ daily history snapshots. ****Effort: Medium.****

Issue 3 — Stochastic Oscillator Missing %D Line Analysis Complete

****Current****: Only Fast %K(14) is calculated. Confirmation threshold: $\text{stochK} < 20$ for oversold, $\text{stochK} > 80$ for overbought.

****What professional platforms use****: Both %K and %D where $\text{\%D} = 3\text{-period SMA of \%K}$. The tradeable signal is the ****%K/%D crossover****, not just %K level.

- ****Interactive Brokers / TradingView****: Display both lines; alert on crossover.
- ****Slow Stochastic**** (industry standard): 3-period SMA of %K creates %D; further smoothing creates the signal line.

****Impact****: The current implementation may flag false confirmations when %K is below 20 but still declining (no reversal yet). A %K cross above %D within the oversold zone is the actual reversal trigger.

****Fix****: Compute ` %D = 3-period SMA of last 3 %K values ` . Add ` StochasticD ` field to ` RsiScanResult ` . ****Effort: Low.****

Issue 4 — RSI Divergence Not Detected Analysis Complete

****Current****: ` CalculateRsiSeries() ` computes the full RSI series (correctly implemented). The ****divergence pattern**** — higher price highs + lower RSI highs = bearish divergence; lower price lows + higher RSI lows = bullish divergence — is not detected algorithmically.

The demo data hardcodes one divergence label (` "Marginal new 52-wk high, but RSI momentum diverging" `), confirming this was conceptually planned but never implemented as a calculation.

****Algorithm approach****:

1. Use the existing ` CalculateRsiSeries() ` output (already available).
2. Find the 2 most recent RSI peaks (overbought) or troughs (oversold) within a 20–60 bar window.
3. Compare with corresponding price peaks/troughs.
4. If price makes new high but RSI peak is lower → bearish divergence.
5. If price makes new low but RSI trough is higher → bullish divergence.

This is a significant signal quality enhancement used by TradingView, StockCharts, and Bloomberg.
****Effort: Medium.****

Issue 5 — MACD EMA Seeding Diverges from TradingView at Startup Analysis Complete

****Current****: ``ema12 = closes.Take(12).Average()`` used as the EMA seed.

****TradingView method****: Seeds the EMA with the ****first closing price**** (not an average), then applies the multiplier from bar 1. This is more consistent with exponential smoothing theory.

****Impact****: Minor divergence in MACD values for the first ~100 bars. With 2 years of daily data (500 bars), this resolves well before the most recent bars. ****Low priority**** for practical use, but worth noting if users compare to TradingView charts.

Issue 6 — Bollinger Band %B and Bandwidth Not Exposed Analysis Complete

****Current****: The calculation returns (upper, middle, lower). ``BollingerBreakout = true/false`` is exposed.

****What is missing****:

- ****%B**** = ``(Price - Lower) / (Upper - Lower)``. Normalises position within the bands (0 = at lower, 1 = at upper, <0 or >1 = outside). More useful than a binary breakout flag.

- ****Bandwidth**** = ``(Upper - Lower) / Middle``. Measures band squeeze. A tight squeeze before a breakout (Bollinger Squeeze) is a key momentum setup used on TradingView, StockCharts, and Bloomberg.

****Fix****: Add ``BollingerPctB`` (0.0–1.0) and ``BollingerBandwidth`` fields to ``RsiScanResult``. Both are computable from values already calculated inside ``CalculateBollingerBands``. ****Effort: Low.****

Issue 7 — Beta Uses Wrong Benchmark for Canadian Stocks Analysis Complete

****Current****: Portfolio beta uses Yahoo Finance's beta field, which measures covariance vs. the ****S&P 500****. For Canadian TSX-listed stocks, this creates a systematic issue because TSX and S&P 500 are not perfectly correlated (~0.80 historically).

****Example****: Royal Bank (RY.TO) has a Yahoo Finance beta of ~0.85 (vs. S&P 500), but vs. the TSX Composite it is ~1.0 (RY is the largest TSX component). The portfolio appears lower-risk than it actually is relative to a Canadian investor's benchmark.

****Sector fallback issue****: ~40% of TSX small/mid-cap symbols have no Yahoo Finance beta → sector proxy used with ±12–20% error.

****What professional Canadian platforms do****: Use TSX Composite (^GSPTSE) as benchmark. Regression beta = $\text{Cov}(\text{Stock}, \text{TSX}) / \text{Var}(\text{TSX})$ from historical returns.

****Recommendation****: Calculate regression beta against TSX Composite using the 2-year daily price data already fetched per symbol in `AnalyzeSymbolAsync``. This eliminates the proxy requirement entirely. ****Effort: Medium.****

Issue 8 — Value Screener F7 (Share Dilution) Uses Wrong Proxy Analysis Complete

****Current****: Piotroski Signal F7 (no new shares issued) is proxied by `RevenueGrowth > 0``. Revenue growth has no relationship to share dilution.

****Correct F7****: `SharesOutstanding_CurrentYear < SharesOutstanding_PreviousYear``. Yahoo Finance `quoteSummary`` returns `sharesOutstanding`` directly.

****Impact****: A company issuing 20% new shares but growing revenue will incorrectly pass F7, inflating its Piotroski score.

****Fix****: Add `SharesOutstanding` to `FundamentalsSnapshot` and compute year-over-year change. ****Effort: Low,****

Issue 9 — Value Screener FCF Fallback Overestimates Capital-Intensive Sectors **Analysis Complete**

****Current****: When real FCF is unavailable, $\text{FCFYield} = \text{OperatingCashFlow} / \text{MarketCap}$.

****Problem****: For pipelines (Enbridge, TC Energy), utilities (Fortis, Emera), and telecoms (BCE, Telus), capital expenditures consume 30–60% of operating cash flow. Using OCF inflates FCF yield by 2–3× for these sectors.

****Example****: Enbridge (ENB.TO) OCF ~\$9B, CapEx ~\$5B → Real FCF ~\$4B. OCF-based yield is 2.25× the real FCF yield.

****Fix****: When `FreeCashFlow` is missing but `CapitalExpenditures` is available, compute $\text{FCF} = \text{OCF} - \text{CapEx}$. Only fall back to pure OCF if both FCF and CapEx are absent. ****Effort: Low,****

Issue 10 — No Dividend Tracking or Total Return **Analysis Complete**

****Current****: Price return only. Dividends received are not tracked.

****Impact****: For TSX dividend payers (banks 4–5%, REITs 5–8%, pipelines 6–8%), ignoring dividends understates total return by 4–8% per year. A position showing +8% price gain may have a +12% total return.

****Industry standard****: Every Canadian broker (TD, RBC, CIBC, Questrade) reports ****total return**** (dividends reinvested) as the primary performance metric.

****Recommendation****:

1. Add a `DividendIncome` table (symbol, amount, date, account).
2. Expose total return = price return + cumulative dividends / original cost.
3. Alternative: integrate Yahoo Finance `dividends` history from `/v8/finance/chart` response (already fetched alongside OHLCV). ****Effort: Medium.****

Issue 11 — No Canadian Adjusted Cost Base (ACB) Tracking Analysis Complete

****Current****: `AverageCostBasis` is a single averaged cost per position. Multiple buy lots are merged.

****Canadian tax law requirement****: The ACB must be tracked precisely for capital gains reporting. It must include all purchase prices, commissions, reinvested dividends, and superficial loss adjustments.

****What competing Canadian tools offer****: Sharesight (AU/CA) and Wealthica (CA) track ACB per lot with full audit trail and capital gains reports by tax year.

****Recommendation****: Track tax lots separately (buy date, price, shares, account type). Full ACB engine requires superficial loss rules (30-day repurchase). ****Effort: High. Priority: High for compliance-conscious Canadian investors.****

Issue 12 — No Options Analytics (Break-Even, Intrinsic Value, Time Value) Analysis Complete

****Current****: Options tracked as positions only. Technical indicators run on the underlying stock. No options-specific analytics exist.

****Simpler immediate wins**** (no option pricing model required):

- ****Break-even price****: $\text{Strike} \pm \text{Premium} / \text{ContractSize}$ — trivial formula.
- ****Intrinsic value****: $\text{Max}(0, \text{CurrentPrice} - \text{Strike})$ for calls, $\text{Max}(0, \text{Strike} - \text{CurrentPrice})$ for puts.
- ****Time value****: $\text{Premium} - \text{IntrinsicValue}$.
- ****Days to expiry****.

****Full Greeks**** (Delta, Gamma, Theta, Vega) require Black-Scholes + implied volatility — either a live IV data source or complex calculation.

****Recommendation****: Add break-even, intrinsic, time value, and days-to-expiry to `OptionItem`. These require no pricing model. ****Effort: Low****

Issue 13 — Currency Risk Not Quantified Analysis Complete

****Current****: CDR symbols (US stocks listed on TSX in CAD) are identified for beta lookups only. No currency allocation or exposure calculation exists.

****Issue****: A portfolio may have 30% USD exposure through CDRs. A 5% CAD weakening vs. USD creates a 1.5% portfolio tailwind — and vice versa. This risk is invisible.

****Recommendation****: Tag positions as CAD/USD exposure. Show currency breakdown in the Allocation screen. ****Effort: Low**** (CDR symbols already identified in beta service).

Issue 14 — EOD Volume Projection Assumes Uniform Distribution Analysis Complete

****Current****: ``ProjectIntradayVolume`` scales raw volume linearly by ``390 / elapsedMinutes``, capped at `2×`.

****Issue****: TSX (and NYSE) volume follows a ****U-curve**** distribution:

- High at open (9:30–10:00 AM): ~20% of daily volume in first 30 minutes.
- Low at midday (11:00 AM–2:00 PM): ~40% over 3 hours.
- High at close (3:30–4:00 PM): ~25% of daily volume in last 30 minutes.

At 3:30 PM (360/390 elapsed), the linear method projects ``1.08×` — but the final 30 minutes typically add 25% of daily volume, so the true projection should be ``~1.33×`. This causes the ``volRatio >= 1.5×` EOD confirm gate to be systematically harder to pass at 3:30–3:45 PM than at 4:00 PM.

****Recommendation****: Use a 30-minute bucket volume profile (13 buckets) with historical TSX distribution weights. ****Effort: Medium.****

Issue 15 — No Position Sizing Framework Analysis Complete

****Current****: Allocation role limits and position limits are configurable percentages. There is no calculation that recommends how many shares to buy given risk parameters.

****Professional platforms provide****:

- **ATR-based sizing** — $\text{Shares} = (\text{Portfolio} \times \text{RiskPercent}) / (\text{EntryPrice} - \text{StopLoss})$. The app already calculates both ATR and dynamic stop loss — this is one formula away.

- **Kelly Criterion** — $f^* = (bp - q) / b$. Theoretically optimal but aggressive.

- **Fixed Fractional** — Risk a fixed % of portfolio per trade.

Recommendation: Expose an ATR-based position size recommendation on the EOD Signals page or as a scanner overlay. $\text{SharesToBuy} = (\text{PortfolioValue} \times \text{MaxRisk\%}) / \text{RiskPerShare}$ where $\text{RiskPerShare} = \text{EntryPrice} - \text{StopLoss}$. **Effort: Low**

Priority & Decision Matrix

#	Improvement	Impact	Effort	Status
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1	TWRR + performance metrics from existing history snapshots — awaiting implementation decision	High	Medium	✓ Analysed
7	Recalculate beta vs. TSX Composite (not S&P 500) — awaiting implementation decision	High	Medium	✓ Analysed
11	Canadian ACB tax lot tracking — awaiting implementation decision	High	High	✓ Analysed
8	Fix Piotroski F7 to use actual shares outstanding — awaiting implementation decision	Medium	Low	✓ Analysed
9	Fix FCF fallback for capital-intensive sectors — awaiting implementation decision	Medium	Low	✓ Analysed
10	Dividend tracking for total return — awaiting implementation decision	High	Medium	✓ Analysed
3	Add Stochastic %D line — awaiting implementation decision	Medium	Low	✓ Analysed
4	RSI divergence detection — awaiting implementation decision	High	Medium	✓ Analysed

12 Options: break-even, intrinsic value, time value awaiting implementation decision	Medium Low  Analysed —
15 ATR-based position sizing recommendation awaiting implementation decision	Medium Low  Analysed —
2 Sharpe/Sortino/Max Drawdown (needs 30+ history days) awaiting implementation decision	High Medium  Analysed —
6 Bollinger %B and Bandwidth implementation decision	Low Low  Analysed — awaiting
13 Currency exposure tracking implementation decision	Low Low  Analysed — awaiting
14 Volume projection curve correction implementation decision	Low Medium  Analysed — awaiting
5 MACD seed alignment with TradingView backlog	Very Low Low  Analysed —